

Python exercises for linguistics

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This is a collection of Python exercises aimed at people in linguistics or natural language processing who are just learning to program. Examples and data are taken from linguistics, and there is a heavy focus on the kinds of tasks that are important when working with language data of different types.

Note: to set up the environment on the triton server (it will not work on your own computer, or the ros server), you need to run the following command *once*:

```
/home/dsv/robe/lis020/bin/setup-lis020
```

This will ensure that editor and ipython settings are identical for all students.

Basic exercises

These exercises can be used to practice the concepts you have learned from each chapter in the Python for Everybody (PY4E) textbook. In general, we recommend that you attempt them in the order presented here. In some cases, exercises build on code from previous exercises, and we have also attempted to put the easier exercises first.

In the PY4E column, you can find a list of the one or two most relevant chapters to the exercise. Of course, further down the list the exercises may use concepts from several earlier chapters, but only the latest chapter(s) are included here.

Some exercises use concepts not included in PY4E. In those cases, the table below contains an asterisk and relevant supplementary readings are provided in the reading list of each exercise.

Exercise	PY4E
<u>Returning the second argument</u>	1, 2, 4
<u>Mandarin tone sandhi</u>	3, 4
<u>Computing bark from frequency</u>	3, 4
<u>Identifying the un-X-able</u>	3, 6
<u>Initial reduplication</u>	3, 6
<u>Swedish verb lemmatizer</u>	3, 6
<u>Words overlap</u>	6, *
<u>Counting syllables</u>	5, 6
<u>Nasals</u>	5, 6
<u>Turkish plurals</u>	5, 6, 8
<u>Censoring text</u>	5, 6, 10
<u>Suffix replacement</u>	5, 9
<u>Finding words of a given part of speech</u>	5, 10
<u>Word by word translation</u>	9
<u>Information density</u>	*
<u>Information density table</u>	3, 5
<u>Colexification of Blood and Die</u>	7
<u>Fingerspelling</u>	7, *
<u>Reaction times</u>	14
<u>Structured documents</u>	14
<u>Raw text files</u>	7, 11, 14

Complex exercises

These exercises do not introduce new Python concepts, but will let you practice applying what you already know to more complex tasks.

- Polish nasal vowels – follow Polish phonological rules and practice string processing and complex conditionals
- Danish to pseudo-Swedish translation – use code from two previous exercises (Word by word translation and Suffix replacement) to practice importing modules and reusing functions.
- Spontal corpus cleaning – filter a real data file in a moderately complex format, and practice handling files as well as getting the logic correct.
- Pokemon name chain – line up Pokemon characters in a name chain, and practice using recursive function calls and managing collections of objects.